Assignment 1

Process Abstractions & Scheduling

Part 1: The Process Abstraction & API

1. **Process State Transitions:** An operating system defines three primary process states: *Running*, *Ready*, and *Blocked*.

   (a) Describe a scenario where a process transitions from *Running* to *Ready*. What OS mechanism forces this transition?

   (b) Describe a scenario where a process transitions from *Running* to *Blocked*. Why doesn't the OS simply leave it in the *Ready* queue?

2. **The Fork-Wait Paradigm:** Analyze the following C code snippet utilizing the POSIX API:

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int x = 100;
    int rc = fork();

    if (rc < 0) {
        fprintf(stderr, "fork failed\n");
    } else if (rc == 0) {
        x += 50;
        printf("Child: x = %d\n", x);
    } else {
        wait(NULL);
        x -= 50;
        printf("Parent: x = %d\n", x);
    }
    return 0;
}
```

   (a) What will be the exact output printed to the terminal?

   (b) Does the `wait(NULL)` system call guarantee the order of the output? Explain your reasoning.

   (c) If the `wait(NULL)` call was removed, how many different valid output orderings could exist?

Part 2: Scheduling Fundamentals

3. **The Convoy Effect & Wait Times:** Consider the following workload of three processes. Assume all processes arrive at $T = 0$, but the OS receives them in the order P_1, P_2, P_3.

- P_1: Burst Time = 100 ms
- P_2: Burst Time = 10 ms
- P_3: Burst Time = 10 ms

(a) Calculate the Average Wait Time and Average Turnaround Time if the OS uses a strict **First Come, First Served (FCFS)** scheduler.

(b) Calculate the Average Wait Time and Average Turnaround Time if the OS uses **Shortest Job First (SJF)**.

(c) Explain the "Convoy Effect" using your mathematical results from parts (a) and (b).

4. **The Fairness Paradox:** You are evaluating two schedulers: **Round Robin (RR)** with a time quantum of 1 ms, and **Shortest Remaining Time First (SRTF)**.

   (a) Prove conceptually why SRTF will always achieve a better (lower) average turnaround time than Round Robin for a mixed workload.

   (b) Despite its mathematical efficiency, why is SRTF considered a dangerous algorithm for interactive, general-purpose operating systems? What specific metric does Round Robin optimize that SRTF destroys?

Part 3: Quantifying Scheduler Performance

While OSTEP defines the core metrics of Turnaround Time and Wait Time, modern operating systems and data center schedulers track advanced statistical metrics to ensure quality of service (QoS).

Advanced Metric Definitions:

- **Makespan:** The total time elapsed from the arrival of the very first process to the completion of the very last process. It measures the absolute throughput of a batch of jobs.
- **P90 Wait Time:** The 90th percentile waiting time. It means that 90% of all processes experienced a wait time less than or equal to this value. It is used to measure the "worst-case" experience, exposing outliers that the average (mean) hides.
- **Jain's Fairness Index ($\mathcal{J}$):** A mathematical measure of how equally a resource is distributed. If x_i is the wait time of process i, and n is the total number of processes, the index is calculated as:

$$\mathcal{J} = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \sum_{i=1}^{n} x_i^2}$$

  The result ranges from $\frac{1}{n}$ (worst-case starvation, only one process gets the resource) to 1.0 (perfect fairness, all processes wait the exact same amount of time).

5. **Calculating the Metrics:** Assume an OS processes a workload of three jobs. The scheduler outputs the following execution trace. (Assume all jobs arrived at $T = 0$).

   - P_1 runs from $T = 0$ to $T = 10$ (Finishes at 10)

- P_2 runs from $T = 10$ to $T = 14$ (Finishes at 14)
- P_3 runs from $T = 14$ to $T = 16$ (Finishes at 16)

(a) Calculate the exact **Wait Time** for P_1, P_2, and P_3.

(b) Calculate the **Makespan** for this workload.

(c) Calculate **Jain's Fairness Index** using the wait times you found in part (a). Show your work. Is this scheduler highly fair or highly unfair?

6. **Analyzing the Outliers (P90):** You are evaluating two different schedulers on a massive workload of 1,000 processes.

- **Scheduler A:** Mean Wait = 15 ms, P90 Wait = 20 ms.
- **Scheduler B:** Mean Wait = 10 ms, P90 Wait = 250 ms.

(a) Which of these two schedulers is highly likely to be utilizing a **Shortest Job First (SJF)** strategy? Explain how the relationship between the Mean and the P90 proves your answer.

(b) If you were deploying a web server that handles user clicks, which scheduler would you choose and why?